AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

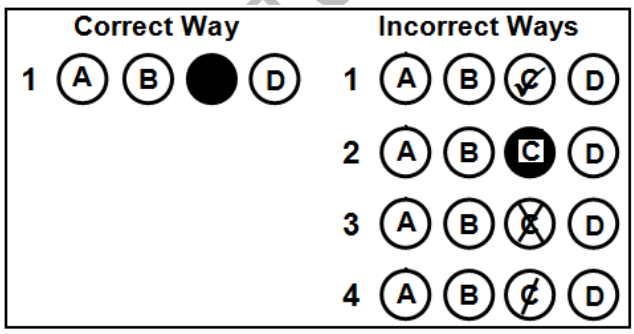
Business Mathematics Paper I

Time: 30 minutes Marks: 20

April/ May 2019 for Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 20 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. A quantity q is increased in the ratio $m : n$, where $m < n$. The new quantity will then be expressed as

   A. $\frac{m}{n} \times q$

   B. $\frac{n}{m} \times q$

   C. $q + \frac{m}{n} \times q$

   D. $q + \frac{n}{m} \times q$

2. The atomic mass of sodium (Na) is 22.98 amu and atomic mass of chlorine (Cl) is 35.5 amu.

   The percentage atomic mass of Na in NaCl is

   A. 154.5
   B. 64.7
   C. 60.7
   D. 39.3

3. The cost price (CP) of an article is Rs $(x + 2)$ and the value of profit or loss is Rs 100.

   The selling price (SP) in terms of x, where SP < CP, is Rs

   A. $x - 98$
   B. $x + 102$
   C. $2x - 100$
   D. $2x + 100$

4. There are 1.5 dozen bananas, 8 apples, 4 pears and 30 apricots in a fruit basket.

   The ratio of bananas to apple in simplest form is

   A. $9 : 4$
   B. $4 : 9$
   C. $8 : 1.5$
   D. $1.5 : 8$

April/ May 2019 for Teaching & Learning only

5. The missing term in the formula of compound interest $A=P(1+?)^{nt}$ is

(**Note**: Symbols have their usual meanings.)

A. $\frac{n}{r}$

B. $\frac{n}{t}$

C. $\frac{r}{n}$

D. $\frac{r}{t}$

6. Ahmed invested Rs 4,000 for 2 years and earned a simple interest of Rs 200. The rate of interest is

A. 7.5%
B. 2.5%
C. 10%
D. 5%

7. In numeral system, octal numbers consist of digits

A. 0, 1 and 7
B. 0, 1, 7 and 8
C. 0, 1, 2, 3, 4, 5, 6 and 7
D. 0, 1, 2, 3, 4, 5, 6, 7 and 8

8. The correct way of converting $(37)_8$ into decimal number system would be

A. $3\times8^1+7\times8^0$

B. $3\times8^0+7\times8^1$

C. $8\times3^0+8\times7^1$

D. $8\times3^1+8\times7^0$

9. $(444)_5 + (134)_5$ is equal to

A. $(223)_5$
B. $(578)_5$
C. $(1023)_5$
D. $(1133)_5$

10. If abscissa of vertex of the parabola $x(x-m)=y$ is -5, then the value of *m* is equal to

A. –10
B. –5
C. 5
D. 10

April/ May 2019 for Teaching & Learning only

11. Which of the following inequalities BEST describes the concavity (concave down) of $a x^2 - 1 = y + 5x^2 + b x$?

A. $a > 5$
B. $a > 0$
C. $a < 0$
D. $a < 5$

12. Which of the following ordered pairs lie(s) on y – axis?

I. (0, 0)
II. (3, 0)
III. (0, 3)

A. I only
B. II only
C. I and III
D. II and III

13. If $px + \frac{p}{2} - 5 = 0$, then the value of $\left(x + \frac{1}{2}\right)$, in terms of p, will be

(**Note**: p is a constant.)

A. $5p$
B. $\frac{5}{p}$
C. $-\frac{5}{p}$
D. $-5p$

14. If $x = by$ and $a = b - 2$, then the value of $\frac{x}{y}$, in terms of a, is equal to

(**Note:** a and b are constants.)

A. $\frac{a}{2}$
B. $2a$
C. $a - 2$
D. $a + 2$

April/ May 2019 for Teaching & Learning only

15. The quadratic equation that has $x = -\frac{1}{k}$ as one of its root is

(**Note**: $k \neq 0$)

A. $\frac{x}{k} - x^2 = 0$

B. $x^2 - \frac{x}{k} = 0$

C. $kx^2 - x = 0$

D. $kx^2 + x = 0$

16. If the difference in roots of the quadratic equation $(2x-3)(4+x)=0$ is a, then the value of a, where $a < 0$, is equal to

A. $-\frac{7}{2}$

B. $-\frac{11}{2}$

C. $\frac{11}{2}$

D. $\frac{7}{2}$

April/ May 2019
for
Teaching & Learning only

Use the given information to answer Q.17, Q.18, Q.19 and Q.20.

Given that $P = \begin{bmatrix} 1 & 0 & -5 \end{bmatrix}$, $Q = \begin{bmatrix} -1 & 0 & -5 \end{bmatrix}$ and $P - R = \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix}^t$,

17. The additive identity of $P - Q$ is

A. $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$
B. $\begin{bmatrix} -2 & 0 & 0 \end{bmatrix}$
C. $\begin{bmatrix} -2 & 0 & 10 \end{bmatrix}$
D. $\begin{bmatrix} 0 & 0 & -10 \end{bmatrix}$

18. The additive inverse of P^t is equal to

A. $\begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$
B. $\begin{bmatrix} -1 \\ 0 \\ -5 \end{bmatrix}$
C. $\begin{bmatrix} -1 & 0 & 5 \end{bmatrix}^t$
D. $\begin{bmatrix} 1 & 0 & -5 \end{bmatrix}^t$

19. The order of matrix *R* will be

A. 1×1
B. 1×3
C. 3×1
D. 3×3

20. The addition of matrices $Q - R$ and $P - Q$

A. is $\begin{bmatrix} 5 & -3 & 7 \end{bmatrix}$
B. is $\begin{bmatrix} 4 & -3 & 2 \end{bmatrix}$
C. is $\begin{bmatrix} 3 & -3 & -3 \end{bmatrix}$
D. cannot be determined.

April/ May 2019 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only